

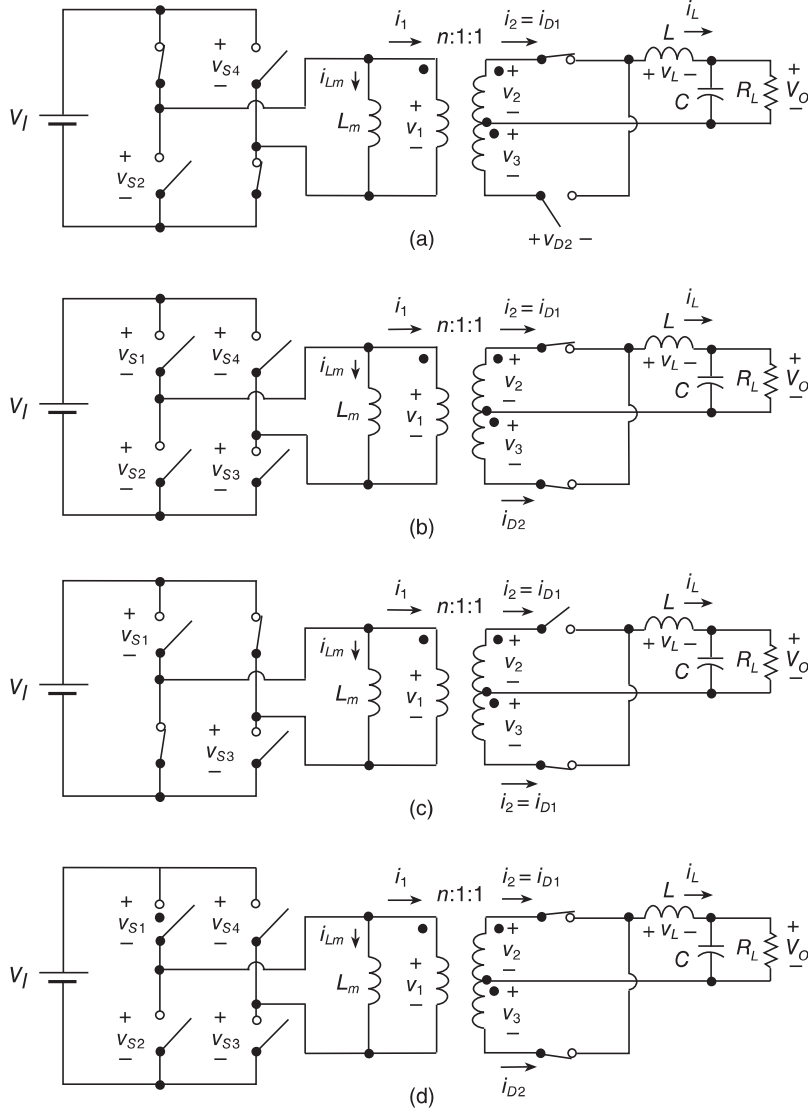


Figure 8.2 Equivalent circuits of the full-bridge converter with a transformer center-tapped rectifier for CCM. (a) For $0 < t \leq DT$. (b) For $DT < t \leq T/2$. (c) For $T/2 < t \leq T/2 + DT$. (d) For $T/2 + DT < t \leq T$.

The voltage across the diode D_2 is

$$v_{D2} = -v_2 - v_3 = -\frac{V_I}{n} - \frac{V_I}{n} = -\frac{2V_I}{n}. \quad (8.10)$$

Since $v_{D2} < 0$, the diode D_2 is OFF. The voltage across the inductor L is given by

$$v_L = \frac{V_I}{n} - V_O = L \frac{di_L}{dt}, \quad (8.11)$$